

Test — COSC 1210 (Computer Science I)

Week of 2026-09-08: variables and statements (ch. 2) and functions I (ch. 3)

Name: _____

- Closed book. No notebook, no laptop, no phone. About 30 minutes.
- Nine questions, one point each.
- Write your answers on this sheet. For “what does it print” parts, write exactly what Python would display. For “write a function” or “write a program” parts, a few lines of Python is enough — we care that it would run and do the right thing, not that the spacing is perfect.
- **Write clearly.** We grade what’s on the page — an answer we can’t read can’t earn credit, even if the code behind it is correct.
- We go over every answer together straight after you hand in.
- Think a question is wrong or badly worded? Say so on the sheet — a good catch earns a bonus point.

1. This program will not run — three of the variable names are illegal. Rewrite the whole thing so that it runs, keeping the meaning of each name.

```
class = "economy"
1st_bag = 20
carry-on = 8
total = 1st_bag + carry-on
```

2. The line below is run with `a = 3` and `b = 4`, so it prints:

```
A 3-4 triangle
```

Rewrite the line so that it uses an **f-string** instead of `+` and `str()` and prints that same text.

```
print("A " + str(a) + "-" + str(b) + " triangle")
```

3. Write a function called `welcome_student` that takes one parameter, `name`, and prints `Welcome, NAME!` using an f-string (so for `name = "Albert"` it prints `Welcome, Albert!`). Then write a line that calls it with the argument `"Albert"`.
4. Write a function called `shout` that takes one parameter, `word`, with a **default value** of `"hello"`, and prints `word` in capital letters (use `.upper()`). Then write two calls: one with no argument at all, one with the argument `"python"`. What does each call print?

5. Here is a function with no docstring:

```
def square(n):  
    print(n * n)
```

- a) Add a docstring to it that says what it does, naming its input and output, the same way we did in class (a one-line summary, then `input:` and `output:` lines).
- b) Which command would only display the docstring?
6. Write a complete function called `total_cost` that takes two parameters, `price` and `quantity`, and prints the total cost (`price` times `quantity`) using an f-string, in the form `Total cost: <amount>`. Give it a docstring.
7. Call your `total_cost` function with `price = 5` and `quantity = 10`.
8. Your `total_cost` function from Q6 does not take a discount. In one sentence, what would you need to add to its parameter list to let a caller optionally apply a percent discount, without breaking any existing call that doesn't pass one?
9. Rewrite `total_cost` so it has a second parameter, `discount`, with a default value of 0, representing a percent discount taken off the total before printing. Call it with `price = 5`, `quantity = 10`, and a discount of 10 percent, and write down what it prints.